

6.7 PreQuiz: Polynomials Challenge — Markscheme

1. [Maximum mark: 7]

Consider the function $f(x) = -2(x+1)(x-3)$, for $x \in \mathbb{R}$.

(a)(i) Write down the x -intercepts. [2]

$x = -1$ and $x = 3$ (A1)(A1)

(a)(ii) Find the coordinates of the vertex. [3]

Axis of symmetry: $x = \frac{-1+3}{2} = 1$ (M1)

$f(1) = -2(1+1)(1-3) = -2(2)(-2) = 8$ (A1)

Vertex: $(1, 8)$ (A1)

(b) $f(x) = -2(x-h)^2 + k$. Write down h and k . [2]

$h = 1$ (A1)

$k = 8$ (A1)

Verification: $-2(x-1)^2 + 8 = -2(x^2 - 2x + 1) + 8 = -2x^2 + 4x - 2 + 8 = -2x^2 + 4x + 6$

And: $-2(x+1)(x-3) = -2(x^2 - 2x - 3) = -2x^2 + 4x + 6 \checkmark$

2. [Maximum mark: 7]

The line $y = mx$ is tangent to the parabola $y = x^2 + 4$.

(a) Show that at any intersection, $x^2 - mx + 4 = 0$. [1]

$x^2 + 4 = mx \implies x^2 - mx + 4 = 0$ (A1)(AG)

(b) Find the discriminant in terms of m . [1]

$\Delta = (-m)^2 - 4(1)(4) = m^2 - 16$ (A1)

(c) Find the values of m for tangency. [2]

Tangent: $\Delta = 0 \implies m^2 - 16 = 0$ (M1)

$m = \pm 4$ (A1)

(d) Find the coordinates of each point of tangency. [3]

For $m = 4$: $x^2 - 4x + 4 = 0 \implies (x-2)^2 = 0 \implies x = 2$ (M1)

$y = 4(2) = 8$. Point of tangency: $(2, 8)$ (A1)

For $m = -4$: $x^2 + 4x + 4 = 0 \implies (x + 2)^2 = 0 \implies x = -2$

$y = -4(-2) = 8$. Point of tangency: $(-2, 8)$ **(A1)**

3. [Maximum mark: 7]

Consider the function $f(x) = 3x^2 - 12x + 7$.

(a) Write $f(x)$ in the form $a(x - h)^2 + k$. [3]

$$f(x) = 3(x^2 - 4x) + 7 \quad (\text{M1})$$

$$= 3(x^2 - 4x + 4 - 4) + 7 = 3(x - 2)^2 - 12 + 7 \quad (\text{M1})$$

$$= 3(x - 2)^2 - 5 \quad (\text{A1})$$

(b) Write down the coordinates of the vertex. [1]

$$\text{Vertex: } (2, -5) \quad (\text{A1})$$

(c) Find the discriminant of $f(x) = 0$. [1]

$$\Delta = (-12)^2 - 4(3)(7) = 144 - 84 = 60 \quad (\text{A1})$$

(d) State the number of x -intercepts and justify. [2]

Since $\Delta = 60 > 0$, $f(x) = 0$ has two distinct real solutions. (A1)

Therefore the graph has two x -intercepts. (R1)

4. [Maximum mark: 8]

Let $p(x) = 2x^3 - x^2 - 13x - 6$.

(a) Show that $(x + 2)$ is a factor of $p(x)$. [2]

$$p(-2) = 2(-8) - (-2)^2 - 13(-2) - 6 \quad (\text{M1})$$

$$= -16 - 4 + 26 - 6 = 0$$

Since $p(-2) = 0$, $(x + 2)$ is a factor by the factor theorem. (A1)(AG)

(b) Express $p(x) = (x + 2)(ax^2 + bx + c)$. [2]

By polynomial division: (M1)

$$2x^3 - x^2 - 13x - 6 = (x + 2)(2x^2 - 5x - 3) \quad (\text{A1})$$

$$\text{i.e. } a = 2, b = -5, c = -3.$$

(c) Factorise $p(x)$ completely. [2]

$$2x^2 - 5x - 3 = (2x + 1)(x - 3) \quad (\text{M1})$$

$$p(x) = (x + 2)(2x + 1)(x - 3) \quad (\text{A1})$$

(d) Write down all solutions to $p(x) = 0$. [1]

$$x = -2, \quad x = -\frac{1}{2}, \quad x = 3 \qquad \textbf{(A1)}$$

(e) Write down the y -intercept. [1]

$p(0) = -6$, so the y -intercept is $(0, -6)$. **(A1)**